

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm Examination University Theory Examination April 2016 PH104 Pharmaceutical Engineering - I

Date: 14/04/2016 , Thursday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following
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|---|---|----|
| A | Discuss material balance and energy balance. | 10 |
| B | Define stoichiometry. Write a short note on different types of graphical representation | 10 |
| C | Discuss Rotameter & Orificemeter in detail | 10 |
- Q 2** Attempt any TWO of the following
- | | | |
|---|--|----|
| A | Describe the various modes of heat transfer. Derive the equation for the rate of heat transfer when the resistances are in series. | 10 |
| B | Discuss the principle involved in Mass transfer. Define Black Body. Write a short note on Stefan-Boltzmann law. | 10 |
| C | 1. Define corrosion. Classify the types of corrosion. How will you prevent corrosion? | 05 |
| | 2. Explain Dalton's Law and Amagat's Law. | 05 |

SECTION- II

- Q 3** Attempt any TWO of the following
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|---|--|----|
| A | 1. What is the importance of packaging for pharmaceutical product? | 5 |
| | 2. Explain the belt conveyor. | 5 |
| B | Explain in detail the pipe system in pharam industry for handing of liquid material. | 10 |
| C | What are the criteria for the selection of material for paln counstruction? | 10 |

Q 4 Attempt any **TWO** of the following

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|---|--|----|
| A | Discuss in detail types of flow. | 10 |
| B | What is the advantages of steam as heating medium? | 10 |
| C | Write a note on glass as packaging material. | 10 |
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